

GPS on Raspberry Pi

SOMNATH BERA

A global positioning system (GPS) module is a device used to determine its location on earth in terms of latitude and longitude. Since Raspberry Pi is a complete

Connecting GPS Module with GPIO of Raspberry Pi

GPS module pin	Raspberry Pi pin
3v3 pin	Pin 2
Gnd pin	Pin 6
Txd pin or Data Out pin	Pin 10
Rxd pin or Data In pin	Pin 8



Fig. 1: GPS module

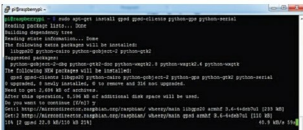


Fig. 2: Installing the GPS libraries



Fig. 3: Load the gpsd

Time: 2013-05-07T17:03:25.000Z	PRN: Elev: Azim: SNR: Used:
Latitude: 24.84669 N	2 48 265 39 Y
Longitude: 82.648226 E	10 54 108 22 Y
Altitude: 304.7 m	4 52 024 23 Y
Speed: 0.1 kph	12 39 323 23 Y
Heading: 344.0 deg (true)	5 13 180 36 Y
CLimb: 0.0 m/min	24 28 255 36 Y
Status: 3D FIX (5 secs)	17 28 072 32 Y
Longitude Err: +/- 2 m	19 21 255 32 Y
Latitude Err: +/- 2 m	28 12 141 38 Y
Altitude Err: +/- 7 m	25 07 322 22 Y
Course Err: n/a	9 05 179 18 N
Speed Err: +/- 16 kph	
Time offset: 0.555	
Grid Square: NL14hc	

Fig. 4: GPS data

computer in itself with a stable operating system therefore connecting a GPS device to it is just like connecting it to any other computer. But getting it to work, then pulling out your chosen GPS-related data in Python and performing some predefined tasks with the GPS data is something very different and interesting.

Circuit and working

There are a variety of GPS modules available in the market but the one with a built-in patch antenna on top (POT) gives the advantage of data reception even indoors. For GPS module to work outdoors, you may have to find a clear sky to receive data. The module can either be connected with the USB cable or with just four wires connected directly to the GPIO of Raspberry Pi as shown in the table.

For simplicity, here we are using the USB cable (refer Fig. 1). After inserting the USB cable look in the /dev/ttyUSB* directory to see which device gets recognised. Most of the times, it becomes /dev/ttyUSB0 but in case we connect it with the GPIO pins, it becomes /dev/ttyAMA0.

Update and install the GPS libraries using below-given commands, also shown in Fig. 2.

```
$ apt-get update
$ apt-get install
gpsd gpsd-clients
python-gps
python-serial
```

Now load the GPS device with the command below, also shown in Fig. 3.

```
$ gpsd /dev/ttyUSB0
-F /var/run/gpsd.sock
```

Write below com-

mand and all the GPS data will start pouring in as shown in Fig. 4. If data does not appear, take the GPS module near the window and try again.

```
$ cat /dev/ttyUSB0
```

Or you can also use the command below for the same.

```
$ cgps -s
```

The 'gpsd' also has a beautiful graphics data output with all the satellite connections as shown in Fig. 5.

Write below-mentioned command for the graphic data output:

```
$ xgps
```

Screenshot of GPS data is shown in Fig. 6.

Square icon in Fig. 5 indicates the WAAS/EGNOS satellite, circles indicate ordinary GPS satellites. Filled icons were used in the last fix of locations.

GPS data takes some time to get synchronised. Therefore wait for some time and if nothing appears, try the steps below:

```
$ sudo killall gpsd
```

```
$ sudo gpsd /dev/ttyUSB0 -F /var/run/gpsd.sock
```

```
$ cat /dev/ttyUSB0
```

The 'gpsd' even has a built-in self telnet daemon at port 2947 with JSON. To use it, first install it in Raspberry Pi using:

```
$ sudo apt-get install telnet
```

Now telnet to local host at port 2947 using below-mentioned command:

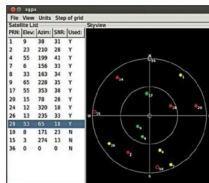


Fig. 5: Satellite connection